

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: DecisionAI

Version: 1.0

1. Project Overview

- **Problem Statement:** Retail SMEs often experience sales decline but lack the tools and analytical expertise to diagnose the root causes. Data is fragmented across transaction logs, financial reports, and external market signals, forcing business owners to rely on intuition or slow manual analysis.
- **Target Domain:** AI-powered decision intelligence for **Retail SMEs (Small & Medium Enterprises)**.
- **Proposed Solution Summary:** DecisionAI is an AI-driven decision intelligence system that enables retail SMEs to upload business data (e.g., CSV financial reports, transaction logs) and receive **instant, structured insights, strategic recommendations, and trade-off analysis**. By leveraging retrieval-augmented generation (RAG), DecisionAI combines internal business data with external market trends to provide **context-aware, explainable decision support**, reducing analysis time from hours to seconds.

2. Background & Business Objective

- **Background of the Problem:** Retail SMEs typically analyze performance manually using spreadsheets or basic dashboards. This approach:
 - Requires time-consuming interpretation
 - Lacks predictive and strategic insights
 - Cannot easily integrate external market context

As a result, businesses struggle to identify **why sales decline occurs** and **what actions to take**.

- **Importance of Solving This Issue:**
 - Faster diagnosis of business problems
 - Reduced reliance on intuition
 - Improved strategic decision-making
 - Increased competitiveness for SMEs

- **Strategic Fit / Impact:** DecisionAi aligns with modern AI-driven analytics trends by:
 - Integrating **LLMs for reasoning and explanation**
 - Using **vector databases (Qdrant)** for contextual retrieval
 - Enabling **data-driven decisions without requiring technical expertise**

3. Product Purpose

3.1. Main Goal of the System: To provide an **AI-powered decision copilot** that helps retail SMEs diagnose business performance issues and recommend actionable strategies.

3.2. Intended Users (Target Audience):

- Retail SME Owner
- Store Managers
- Business Analyst (SME Level)
- Entrepreneurs

- Frontend Developers
- Low Code Builders

4. System Functionalities

4.1. Description: The system operates as a decision intelligence engine:

1. Users answer guided questions specifically tailored to gain as much information as possible for analysis and insight generation
2. Optionally choose to upload structured data (CSV files) , pick from already available data or upload no data at all.
3. System processes and embeds data into a knowledge base
4. AI retrieves relevant context
5. AI generates structured insights, recommendations, and predictions

4.2. Key Functionalities:

- **Data Ingestion & Understanding:**
 - Accepts CSV uploads (sales, transactions, financial reports)
 - Converts structured, unstructured data into embeddings
- **Contextual Retrieval:** Uses vector search (Qdrant) to retrieve relevant historical + market data.
- **AI Decision Engine:** Generates:
 - Insights
 - Recommendation
 - Predictions
 - Trade-off analysis
- **Explainable Output:** Structured JSON Output containing:
 - Insight
 - Reasoning

- Recommendations
- Prediction
- Trade-offs (option A, Option B, final verdict)

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Embedding model used for RAG is gemini-embedding-2, and LLM used is gemini-2.5-flash. The selected AI have strong reasoning capabilities, support JSON schema, efficient for real-time decision systems, and can work together as they are both from Google Gemini.

4.3.2. Prompting Strategy: Multi-step RAG prompting

- Step 1: Retrieve relevant context
- Step 2: Inject into prompt
- Step 3: Generate structured decision output

These Steps Ensure context-awareness reasoning, reduces hallucinations, and improves accuracy of recommendations.

4.3.3. Context & Input Handling:

- Input Types:
 - CSV structured
 - User Input (unstructured)
- Handling Strategy:
 - Inputs are embedded using Gemini embedding models and stored in Qdrant
 - Vectors are retrieved via top-k similarity search

- Max Input:
 - Limited by embedding + LLM token constraints

4.3.4. Fallback & Failure Behavior:

- Structured schema validation prevents malformed outputs
- If response is unusable:
 - Users can go back to previous steps to refine their inputs

5. User Stories & Use Cases

- **User Stories:**
 - As a retail business owner, I want to upload my sales data so that I can understand why my revenue is declining.
 - As a manager, I want actionable recommendations so I can improve performance quickly.
- **Use Cases (Main Interactions):**

1. Sales Diagnosis Flow:

1. User uploads CSV
2. System processes data
3. AI generates insights:
 - Root causes of decline
 - Key trends
 - Risk factors

2. Strategy Recommendation: Ai suggest:

- Price changes
- Inventory adjustments
- Marketing strategies

3. Trade-off Analysis: Ai provides two options and explains:

- Pros vs cons
- Risk levels
- Expected outcomes

6. Features Included (Scope Definition)

- CSV data ingestion
- RAG-based contextual retrieval
- AI-generated structured insights
- Trade-off analysis engine
- Prediction generation
- Explainable reasoning outputs

7. Features Not Included (Scope Control)

- Real-time POS system integration
- Multi-user collaboration
- Full financial forecasting models
- General AI usage for multiple industries

8. Assumptions & Constraints

- **LLM Cost Constraint:** Currently there is no cost as all the models used are free models. To optimize token usage, we used RAG to reduce prompt size and provide example datasets already embedded in database for repeated use.
- **Technical Constraints:** Currently is only limited to CSV input format for input files.
- **Performance Constraints:** Large datasets may increase latency and there is a token limit thus restraining full dataset integration.
- **User Input:** Requires meaningful user data input and output quality depends on data quality.

9. Risks & Questions Throughout Development

- **Data Quality Risk:** Poor or incomplete data may lead to inaccurate insights/ predictions
- **Model Hallucination:** Mitigated using RAG context injection, structured output schema, and guided user input
- **Variable Interpretations:** Users may misinterpret recommendations